

Probability Theory and Statistics

Exercise 10 solution

11.10. – 11.14.

Conditional expectation, law of total probability in continuous case, law of total expectation

1. As a sample space we may take the Cartesian product $[6]^3 = \{1, 2, 3, 4, 5, 6\}^3$ with the classical probability measure, so each outcome has probability $\frac{1}{216}$. By the definition of conditional expectation on an event,

$$\mathbb{E}(X \mid Y = 3) = \sum_{k \in \text{Ran}(X)} k \mathbb{P}(X = k \mid Y = 3) = \sum_{k \in \text{Ran}(X)} k \frac{\mathbb{P}(X = k, Y = 3)}{\mathbb{P}(Y = 3)}.$$

Here

$$\mathbb{P}(Y = 3) = \mathbb{P}(\text{all three outcomes} \leq 3) - \mathbb{P}(\text{all three outcomes} \leq 2) = \frac{3^3}{6^3} - \frac{2^3}{6^3} = \frac{19}{216}.$$

Since X is the minimum and Y the maximum of the three throws, we have $\mathbb{P}(X = k, Y = 3) = 0$ for $k > 3$. Moreover,

$$\mathbb{P}(X = 3, Y = 3) = \mathbb{P}(\text{all three are } 3) = \frac{1}{216},$$

$$\mathbb{P}(X = 2, Y = 3) = \mathbb{P}(\{(2, 2, 3), (2, 3, 2), (3, 2, 2), (3, 3, 2), (3, 2, 3), (2, 3, 3)\}) = \frac{6}{216},$$

and

$$\mathbb{P}(X = 1, Y = 3) = \frac{12}{216}.$$

(Alternatively: $\mathbb{P}(X = 1, Y = 3) = \mathbb{P}(Y = 3) - \mathbb{P}(X = 2, Y = 3) - \mathbb{P}(X = 3, Y = 3) = \frac{12}{216}$.)

Hence

$$\mathbb{E}(X \mid Y = 3) = 1 \cdot \frac{12/216}{19/216} + 2 \cdot \frac{6/216}{19/216} + 3 \cdot \frac{1/216}{19/216} = \frac{12}{19} + \frac{12}{19} + \frac{3}{19} = \frac{27}{19} = 1 \frac{8}{19}.$$

2. By the definition of conditional expectation on an event,

$$\mathbb{E}(Y \mid A) = \sum_{k \in \text{Ran}(Y)} k \mathbb{P}(Y = k \mid A) = \sum_{k \in \text{Ran}(Y)} k \frac{\mathbb{P}(\{Y = k\} \cap A)}{\mathbb{P}(A)}.$$

Independence of tosses gives

$$\mathbb{P}(A) = \mathbb{P}(\text{1st toss is T, 2nd is the first H}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Also $\mathbb{P}(\{Y = 4\} \cap A) = \mathbb{P}(\{Y = 0\} \cap A) = 0$ since under A the total number of heads cannot be 0 or 4. We can model the sample space by all length-4 sequences of H/T, which has 16 equiprobable outcomes. Then

$$\mathbb{P}(\{Y = 1\} \cap A) = \mathbb{P}(\text{only the 2nd toss is H}) = \frac{1}{16},$$

$$\mathbb{P}(\{Y = 2\} \cap A) = \mathbb{P}(\text{1st is T, 2nd is H, and exactly one of 3rd/4th is H}) = \frac{2}{16},$$

$$\mathbb{P}(\{Y = 3\} \cap A) = \mathbb{P}(\text{1st is T, the rest are H}) = \frac{1}{16}.$$

Therefore

$$\mathbb{E}(Y \mid A) = 1 \cdot \frac{1/16}{1/4} + 2 \cdot \frac{2/16}{1/4} + 3 \cdot \frac{1/16}{1/4} = 2.$$

Remarks.

- One can also use $\mathbb{E}(Y \mid A) = \frac{\mathbb{E}(Y \mathbf{1}_A)}{\mathbb{P}(A)}$.
 - Alternatively, under A the number of heads is 1 (from the 2nd toss) plus a $\text{Bin}(n = 2, p = 1/2)$ variable (heads among 3rd and 4th), whose mean is 1, giving 2.
3. We obtain $\mathbb{E}(Y \mid X)$ by plugging X into the regression function $g(k) = \mathbb{E}(Y \mid X = k)$. Fix $k \in \{0, 1, \dots, n\}$. Given $X = k$, there are k sixes (all even), and $n - k$ non-sixes. For a single die Z ,

$$\mathbb{P}(Z = j \mid Z \neq 6) = \frac{1/6}{5/6} = \frac{1}{5} \quad (j = 1, \dots, 5), \quad \mathbb{P}(Z \text{ even} \mid Z \neq 6) = \frac{2}{5}.$$

By independence, conditional on the k sixes, among the remaining $n - k$ throws the number of even outcomes is $\text{Bin}(n - k, 2/5)$ whose mean is $\frac{2}{5}(n - k)$. Hence

$$g(k) = \mathbb{E}(Y \mid X = k) = k + (n - k) \frac{2}{5} = \frac{3}{5}k + \frac{2}{5}n, \quad \text{so} \quad \mathbb{E}(Y \mid X) = \frac{3}{5}X + \frac{2}{5}n.$$

4. (a) Let $x \in (1, 2)$. Given $X = x$, we have $Y \sim \text{Exp}(x)$. Thus

$$g(x) := \mathbb{E}(Y \mid X = x) = \int_0^\infty y x e^{-xy} dy = \frac{1}{x}.$$

Therefore $\mathbb{E}(Y \mid X) = g(X) = \frac{1}{X}$.

(b) By the definition of conditional density, for $x \in (1, 2)$,

$$f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y \mid x) = 1 \cdot (x e^{-xy} \mathbf{1}_{\{y > 0\}}).$$

Outside $(1, 2)$ in x or for $y \leq 0$, the joint density is 0. In summary,

$$f_{X,Y}(x, y) = \begin{cases} x e^{-xy}, & 1 < x < 2, \ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

5. (a) By definition, $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$ for $f_X(x) > 0$, and 0 otherwise. For $0 < x < 1$,

$$f_X(x) = \int_0^1 \frac{12}{5} (x^2 - xy + y^2) dy = \frac{12}{5} \left(x^2 - \frac{x}{2} + \frac{1}{3} \right).$$

Hence, for $x \in (0, 1)$ and $y \in (0, 1)$,

$$f_{Y|X}(y \mid x) = \frac{\frac{12}{5} (x^2 - xy + y^2)}{\frac{12}{5} \left(x^2 - \frac{x}{2} + \frac{1}{3} \right)} = \frac{6(x^2 - xy + y^2)}{6x^2 - 3x + 2},$$

and $f_{Y|X}(y | x) = 0$ otherwise. Summarizing,

$$f_{Y|X}(y | x) = \begin{cases} \frac{6(x^2 - xy + y^2)}{6x^2 - 3x + 2}, & x, y \in (0, 1), \\ 0, & \text{otherwise.} \end{cases}$$

(b) Using the regression formula for the continuous case,

$$g(x) := \mathbb{E}(Y | X = x) = \int_0^1 y \frac{6(x^2 - xy + y^2)}{6x^2 - 3x + 2} dy = \frac{3x^2 - 2x + \frac{3}{2}}{6x^2 - 3x + 2}.$$

Thus

$$\mathbb{E}(Y | X) = \frac{3X^2 - 2X + \frac{3}{2}}{6X^2 - 3X + 2}.$$

(c) For the covariance matrix, compute $\text{Var}(X)$, $\text{Var}(Y)$ and $\text{cov}(X, Y)$.

$$\mathbb{E}(X) = \int_0^1 x f_X(x) dx = \frac{12}{5} \int_0^1 \left(x^3 - \frac{x^2}{2} + \frac{x}{3} \right) dx = \frac{3}{5},$$

$$\mathbb{E}(X^2) = \int_0^1 x^2 f_X(x) dx = \frac{12}{5} \int_0^1 \left(x^4 - \frac{x^3}{2} + \frac{x^2}{3} \right) dx = \frac{67}{150},$$

so $\text{Var}(X) = \frac{67}{150} - \left(\frac{3}{5}\right)^2 = \frac{13}{150}$. By symmetry, $\text{Var}(Y) = \frac{13}{150}$ as well.

Next,

$$\mathbb{E}(XY) = \int_0^1 \int_0^1 xy \frac{12}{5} (x^2 - xy + y^2) dx dy = \frac{1}{3},$$

hence $\text{cov}(X, Y) = \frac{1}{3} - \left(\frac{3}{5}\right)^2 = -\frac{2}{75}$. Therefore the covariance matrix is

$$\begin{pmatrix} \text{Var}(X) & \text{cov}(X, Y) \\ \text{cov}(X, Y) & \text{Var}(Y) \end{pmatrix} = \begin{pmatrix} \frac{13}{150} & -\frac{2}{75} \\ -\frac{2}{75} & \frac{13}{150} \end{pmatrix}.$$

(d) The linear regression parameters are

$$\beta = \frac{\text{cov}(X, Y)}{\text{Var}(X)} = \frac{-2/75}{13/150} = -\frac{4}{13}, \quad \alpha = \mathbb{E}(Y) - \beta \mathbb{E}(X) = \frac{3}{5} + \frac{4}{13} \cdot \frac{3}{5} = \frac{51}{65}.$$

Thus the linear regression of Y on X is $-\frac{4}{13}X + \frac{51}{65}$.

6. Fix $y \in \text{Ran}(Y)$ and compute $\mathbb{E}(X | Y = y)$. By the lecture formula,

$$\mathbb{E}(X | Y = y) = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx, \quad f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

First find the marginal of Y :

$$f_Y(y) = \int_0^1 \frac{4}{5} (x + y + xy) dx = \frac{4}{5} \left(\frac{1}{2} + y + \frac{1}{2}y \right) = \frac{6}{5}y + \frac{2}{5}.$$

Hence

$$f_{X|Y}(x | y) = \frac{\frac{4}{5}(x + y + xy)}{\frac{6}{5}y + \frac{2}{5}} = \frac{2(x + y + xy)}{3y + 1}.$$

Therefore

$$\mathbb{E}(X \mid Y = y) = \int_0^1 x \frac{2(x + y + xy)}{3y + 1} dx = \frac{2}{3y + 1} \int_0^1 (x^2 + xy + x^2y) dx = \frac{5y + 2}{9y + 3},$$

so the regression is $\frac{5Y + 2}{9Y + 3}$.

7. (a) (10 points)

(1 pt) $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)},$

(1 pt) valid when $f_X(x) > 0$ (we return to the case $f_X(x) = 0$ later).

(1 pt) First find $f_X(x)$; $f_X(x) = 0$ if $x \leq 1$ or $x \geq 3$,

(1 pt) and if $1 < x < 3$, then $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$

(1 pt) $= \int_2^{2x} \frac{1}{4} dy$ (it suffices to write this, provided $1 < x < 3$ is clear)

(1 pt) $= \left[\frac{1}{4} y \right]_{y=2}^{y=2x}$

(1 pt) $= x/2 - 1/2.$

(1 pt) Thus, for $1 < x < 3$: if $2 < y < 2x$, then $f_{Y|X}(y \mid x) = \frac{1/4}{x/2 - 1/2} = \frac{1}{2x - 2},$

(1 pt) and (still for $1 < x < 3$) if $y \leq 2$ or $y \geq 2x$, then $f_{Y|X}(y \mid x) = 0.$

(1 pt) Moreover, $f_{Y|X}(y \mid x) = 0$ for all y if $f_X(x) = 0$, i.e., $x \leq 1$ or $x \geq 3.$

(b) (6 points)

(1 pt) The regression function is $g(x) = \mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x) dy,$

(1 pt) $= \int_2^{2x} \frac{y}{2x - 2} dy$

(0 pt) $= \frac{1}{2x - 2} \int_2^{2x} y dy$

(1 pt) $= \frac{1}{2x - 2} \left[\frac{y^2}{2} \right]_{y=2}^{y=2x}$

(1 pt) $= \frac{1}{2x - 2} (2x^2 - 2)$

(1 pt) $= \frac{(x-1)(x+1)}{x-1} = x + 1.$

(1 pt) Hence $\mathbb{E}(Y \mid X) = g(X) = X + 1.$

Second solution.

(1 pt) From part (a), for $1 < x < 3$ the function $y \mapsto f_{Y|X}(y \mid x)$ is constant on $(2, 2x)$ (and 0 otherwise),

(2 pts) hence given $X = x$, Y is uniform on $(2, 2x)$ (equivalently, given $X, Y \mid X$ is $U(2, 2X)$),

(2 pts) so $\mathbb{E}[Y \mid X = x]$ equals the midpoint of $(2, 2x)$, i.e., $\frac{2+2x}{2} = x + 1,$

(1 pt) thus $\mathbb{E}[Y \mid X] = X + 1.$

(c) (4 points)

First solution.

(1+1 pts) Using f_X from (a), $\mathbb{P}(X < 2) = \int_{-\infty}^2 f_X(x) dx = \int_1^2 \left(\frac{x}{2} - \frac{1}{2} \right) dx$

(1 pt) $= \left[\frac{x^2}{4} - \frac{x}{2} \right]_1^2$

(1 pt) $= 1 - 1 - \frac{1}{4} + \frac{1}{2} = \frac{1}{4}.$

Second solution (direct from the joint density).

(1 pt) $\mathbb{P}(X < 2) = \int_1^2 \int_2^{2x} \frac{1}{4} dy dx$

$$(1 \text{ pt}) = \int_1^2 \left[\frac{1}{4}y \right]_{y=2}^{2x} dx$$

$$(1 \text{ pt}) = \int_1^2 \left(\frac{1}{2}x - \frac{1}{2} \right) dx$$

$$(1 \text{ pt}) = \left[\frac{1}{4}x^2 - \frac{1}{2}x \right]_1^2 = \frac{1}{4}.$$

Third solution (geometric probability).

(1 pt) (X, Y) is uniformly distributed over the right triangle

$$\Omega = \{(x, y) \in \mathbb{R}^2 : 1 < x < 3, 2 < y < 2x\},$$

whose area is 4.

(1 pt) The favorable region is the portion of Ω with $x < 2$, a similar right triangle of legs 1 and 2, so its area is 1.

(1 pt) Therefore $\mathbb{P}(X < 2) = \frac{\text{favorable area}}{\text{total area}} = \frac{1}{4}$.

8. The joint density $f_{X,Y}$ and the marginal f_X were computed on Sheet 8. Thus

$$f_{Y|X}(y | x) = 0 \quad \text{if } f_X(x) = 0 \quad (x \leq 0 \text{ or } x \geq \pi),$$

and for $0 < x < \pi$,

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{1}{\sin x}.$$

In summary,

$$f_{Y|X}(y | x) = \begin{cases} \frac{1}{\sin x}, & 0 < x < \pi \text{ and } 0 < y < \sin x, \\ 0, & \text{otherwise.} \end{cases}$$

The conditional distribution function is obtained by integrating in y ; where $f_X(x) = 0$ we may set it to 0:

$$F_{Y|X}(y | x) = \int_{-\infty}^y f_{Y|X}(t | x) dt = \begin{cases} 0, & x \leq 0 \text{ or } x \geq \pi \text{ or } (0 < x < \pi \text{ and } y \leq 0), \\ \frac{y}{\sin x}, & 0 < x < \pi \text{ and } 0 < y < \sin x, \\ 1, & 0 < x < \pi \text{ and } y \geq \sin x. \end{cases}$$

9. (a) By linearity of conditional expectation and independence,

$$\mathbb{E}(3X - Y + 1 | X) = 3\mathbb{E}(X | X) - \mathbb{E}(Y | X) + 1 = 3X - \mathbb{E}(Y) + 1 = 3X + \frac{1}{2}.$$

(b)

$$\mathbb{E}((2XY)^2 - 7Y | X) = 4X^2 \mathbb{E}(Y^2 | X) - 7\mathbb{E}(Y | X) = 4X^2 \mathbb{E}(Y^2) - 7\mathbb{E}(Y).$$

For $Y \sim \text{Exp}(2)$, $\mathbb{E}(Y) = \frac{1}{2}$ and $\text{Var}(Y) = \frac{1}{4}$, hence $\mathbb{E}(Y^2) = \text{Var}(Y) + (\mathbb{E}Y)^2 = \frac{1}{2}$. Thus the result is

$$2X^2 - \frac{7}{2}.$$

(c)

$$\mathbb{E}(X^2 + 2XY + Y^2 \mid X + Y) = \mathbb{E}((X + Y)^2 \mid X + Y) = (X + Y)^2.$$

(d)

$$\mathbb{E}\left(X^2 \tan Y + \frac{5X}{Y} - 2Y \mid Y\right) = \tan(Y) \mathbb{E}(X^2 \mid Y) + \frac{5}{Y} \mathbb{E}(X \mid Y) - 2 \mathbb{E}(Y \mid Y).$$

By independence, $\mathbb{E}(X \mid Y) = \mathbb{E}(X) = 0$ and $\mathbb{E}(X^2 \mid Y) = \mathbb{E}(X^2) = 1$ for $X \sim N(0, 1)$. Hence the result is

$$\tan Y - 2Y.$$

10. (2 pts) $\mathbb{E}(Z \mid Y) = \mathbb{E}(2X \ln Y + X^2 Y^2 - 3Y \mid Y)$

(1 pt) (linearity of conditional expectation)

(2 pts) $= \ln Y \cdot \mathbb{E}(2X \mid Y) + Y^2 \cdot \mathbb{E}(X^2 \mid Y) + \mathbb{E}(-3Y \mid Y)$

(2 pts) („taking out what is known”: functions of Y factor out)

(2 pts) $\mathbb{E}(-3Y \mid Y) = -3Y$

(2 pts) and $\mathbb{E}(2X \mid Y) = \mathbb{E}(2X)$, $\mathbb{E}(X^2 \mid Y) = \mathbb{E}(X^2)$ (independence).

(1 pt) With $X \sim N(2, 4)$, $\mathbb{E}(X^2) = \text{Var}(X) + (\mathbb{E}X)^2 = 4 + 2^2 = 8$.

(3 pts) Therefore

$$\mathbb{E}(Z \mid Y) = 4 \ln Y + 8Y^2 - 3Y.$$

11. (a) Let Y be the time to find the file, and $A = \{\text{the file is on the first drive}\}$. Using the law of total expectation over the partition $\{A, \bar{A}\}$,

$$\mathbb{E}(Y) = \mathbb{E}(Y \mid A)\mathbb{P}(A) + \mathbb{E}(Y \mid \bar{A})\mathbb{P}(\bar{A}).$$

With $\mathbb{P}(A) = 2/3$ and $\mathbb{P}(\bar{A}) = 1/3$, we have $\mathbb{E}(Y \mid A) = 7$ and $\mathbb{E}(Y \mid \bar{A}) = 10$ (both drives must be scanned if the file is not on the first). Hence

$$\mathbb{E}(Y) = 7 \cdot \frac{2}{3} + 10 \cdot \frac{1}{3} = 8 \text{ minutes.}$$

(b) Now let Z be the time to find the file when starting with the second drive (same A). Then $\mathbb{E}(Z \mid \bar{A}) = 3$ and $\mathbb{E}(Z \mid A) = 10$. Thus

$$\mathbb{E}(Z) = 3 \cdot \frac{1}{3} + 10 \cdot \frac{2}{3} = \frac{23}{3} = 7\frac{2}{3} \text{ minutes.}$$

12. Let X be the time to exit the mine. Define A_i as the event of choosing path i on a decision, with $\mathbb{P}(A_i) = \frac{1}{3}$ for $i = 1, 2, 3$. We have $\mathbb{E}(X \mid A_1) = 1$, $\mathbb{E}(X \mid A_2) = \mathbb{E}(X) + 2$, and $\mathbb{E}(X \mid A_3) = \mathbb{E}(X) + 3$. By the law of total expectation,

$$\mathbb{E}(X) = \sum_{i=1}^3 \mathbb{E}(X \mid A_i) \mathbb{P}(A_i) = \frac{1}{3} (1 + (\mathbb{E}X + 2) + (\mathbb{E}X + 3)).$$

Solving gives $\mathbb{E}(X) = 6$.